Peihong Yuan

Curriculum Vitae

198 Sherman St, FL1
02140 Cambridge, Massachusetts
+1(617)599-5738
phyuan@mit.edu
Google Scholar

Short Biography

Peihong Yuan received his M.Sc. and Dr.-Ing. degrees in electrical and computer engineering from the Technical University of Munich (TUM) in 2015 and 2021, respectively. Since 2022, he has been a postdoctoral associate in the Research Laboratory of Electronics (RLE) at the Massachusetts Institute of Technology (MIT). During his graduate studies, he served as a lecturer at TUM Asia, Singapore, in 2021 and as an invited lecturer at Tongji University, China, in 2019, 2020, and 2021. He served as a Guest Editor for Entropy-Special Issues: Advances in Modern Channel Coding in 2025 and as a TPC member for the IEEE International Symposium on Topics in Coding (ISTC) 2025, IEEE GlobeCom 2024: Workshop on Channel Coding Beyond 5G, and the IEEE Vehicular Technology Conference 2023. He was recognized as an Exemplary Reviewer for the IEEE Transactions on Communications in 2022. In 2019, he received the Heinrich Hecht Award for winning the coding challenge, "Best Channel Codes for Short Underwater Messages". His research interests include applied probability, information theory, and coding theory.

Education

2015.11–2021.11 **Dr.-Ing. in Electrical and Computer Engineering**, *Technical University of Munich, Germany*.

- Thesis: Polar Coding with Complexity-Adaptive Decoding and Time-Varying Channels
  Supervised by Prof. Gerhard Kramer

2012.9–2015.9 **M.Sc. in Electrical and Computer Engineering**, *Technical University of Munich, Germany & Tongji University, China*.

- Focus on coding theory, communication and information theory
- Thesis: Rate-matched coded modulation in wireless communication
  Supervised by Dr. Georg Böcherer, Patrick Schulte, Markus Stinner

2007.9–2011.7 **B.Sc. in Transportation Engineering**, *Tongji University, China*.

Research Experience

2022.2– **Postdoctoral Associate**, *Massachusetts Institute of Technology, MA, USA*.

Network Coding and Reliable Communications (NCRC) Group (Prof. Muriel Médard)

- Error correction codes for ultra-reliable low-latency communications (DARPA project)
- Joint decoding of random linear network codes (RLNCs) and error correction codes
- Secure and efficient network coding for wireless networks
- Soft-output decoding algorithms

2015.11–2022.2 **Doctoral Researcher**, *Technical University of Munich, Germany*.
Institute for Communications Engineering (LNT)

- Complexity-adaptive (near-)optimum decoding algorithms for polar (and polar-like) codes
- Non-binary polar coding and higher-order message passing
- Polar-coded channel estimation and hybrid ARQ
- Hybrid ARQ for probabilistic amplitude shaping
- Wiretap channel and information-theoretic security
- Polar & LDPC decoders on Xilinx FPGA

2015.2–2015.9 **Master Thesis Student**, *Technical University of Munich, Germany*.
Institute for Communications Engineering (LNT)

- Turbo coding and high performance decoding for probabilistic amplitude shaping
- Type check for constant composition distribution matching

Teaching Experience

Lecturer

2023 **Discrete Mathematics and Proof for Computer Science**, *Massachusetts Institute of Technology, USA*, Undergraduate course 6.120A.

2021 **Digital Communications**, *TUM Asia, Singapore*, 90-hour/3-week Intensive Course (Undergraduate) EDE 3205.

2019-2021 **Aspects of Machine Learning for Communications**, *Tongji University, China*, 35-hour/1.5-week Invited Intensive Course (Graduate).

Lab Supervisor

2019-2021 **Wireless Communications Laboratory**, *Technical University of Munich*.

2016-2020 **Communications Laboratory**, *Technical University of Munich*.

Teaching Assistant

2018-2021 **System Aspects in Communications**, *Technical University of Munich*.

2018-2021 **Coded Modulation**, *Technical University of Munich*.

2017, 2020 **Machine Learning for Communications**, *Technical University of Munich*.

2016 **Advanced Topics in Signal Processing**, *Technical University of Munich*.

Thesis Supervision

2015-2021 **Bachelor & Master thesis**, Supervision of more than 20 Bachelor (undergraduate) and Master (graduate) students at Technical University of Munich..

Research Visits

Mar. 2018 Hosted by Prof. Haim Permuter at *Ben-Gurion University of the Negev, Israel*.

Dec. 2018 Hosted by Prof. Ido Tal at *Technion - Israel Institute of Technology, Israel*.

Professional Service

2025 Guest Editor for Entropy - Special Issues: Advances in Modern Channel Coding.

2025 Technical Program Committee (TPC) Member for IEEE International Symposium on Topics in Coding (ISTC).

2024 Technical Program Committee (TPC) Member for IEEE GlobeCom: Workshop on Channel Coding beyond 5G.

2023 Technical Program Committee (TPC) Member for IEEE Vehicular Technology Conference - Spring.

Accomplishments

2022 Exemplary Reviewer for the IEEE Transactions on Communications

2019 Heinrich Hecht Award (for Ultra-Reliable Low-Latency Code Design).

2014-2015 DAAD scholarship holder during the master education.

2012-2014 National scholarship for outstanding student in Tongji university

2013 Huate academic scholarship (best 25%)

Selected Publications

Journal Papers (submitted)

1 J. Griffin, **P. Yuan**, R. Thesmar, P. Popovski, K. R. Duffy, M. Médard, "Code at the Receiver, Decode at the Sender: Feedback Communication with GRAND-CE," submitted to IEEE Trans. Commun., pages 1-11, 2023. (major revision)

2 H. Sarieddeen, **P. Yuan**, M. Médard, K. R. Duffy, "Soft-input, Soft-output Turbo-GRAND," submitted to IEEE Trans. Signal Process., pages 1-11, 2022. (major revision)

3 **P. Yuan**, K. R. Duffy, E. P. Gabhart, M. Médard, "On the Role of Quantization of Soft Information in GRAND," submitted to IEEE Trans. Signal Process., pages 1-13, 2022. (major revision)

Journal Papers (published)

4 **P. Yuan**, M. Médard, K. Galligan, K. R. Duffy, "Soft-output (SO) GRAND and Iterative Decoding to Outperform LDPC Codes," IEEE Trans. Wirel. Commun., 2025. (accepted)

5 **P. Yuan**, K. R. Duffy, M. Médard, "Soft-Output Successive Cancellation List Decoding," IEEE Trans. Inf. Theory, 2024.

6 K. R. Duffy, **P. Yuan**, J. Griffin, M. Médard, "Soft-output Guessing Codeword Decoding," IEEE Commun. Letters, 2024.

7 L. Shen, Y. Wu, **P. Yuan**, C. Xiao, X. Xia, W. Zhang, "Probabilistic Shaped Multilevel Polar Coding for Wiretap Channel," IEEE J. Sel. Areas Commun., 2024. (accepted)

8 **P. Yuan**, M. Coşkun, "Successive Cancellation Ordered Search Decoding of Modified GN-coset Codes," IEEE Trans. Commun., 2024.

9 B. Shi, Y. Wu, **P. Yuan**, D. W. K. Ng, X. -G. Xia, W. Zhang, "Code-Aided Channel Estimation in LDPC-Coded MIMO Systems," IEEE Wirel. Commun. Letters, 2023

10 **P. Yuan**, M. Coşkun, G. Kramer, "Polar-coded non-coherent communication," IEEE Commun. Letters, 2021

11 T. Wiegart, F. Steiner, P. Schulte, **P. Yuan**, "Shaped on-off keying using polar codes," IEEE Commun. Letters, 2019

Conference Papers

1 **P. Yuan**, K. R. Duffy, M. Médard, "Near-Optimal Generalized Decoding of Polar-like Codes," ISIT, 2024.

2 **P. Yuan**, K. R. Duffy, M. Médard, "Time-Shift Coding for Uncoordinated MACs," LatinCom, 2023.

3 **P. Yuan**, K. R. Duffy, E. Gabhart, M. Médard, "Guessing Random Additive Noise Decoding with Quantized Soft Information," GlobeCom Workshop, 2023.

4 K. Galligan, **P. Yuan**, M. Médard, K. R. Duffy, "Upgrade error detection to prediction with GRAND," GlobeCom, 2023.

5 K. R. Duffy, M. Médard, **P. Yuan**, "Homomorphic Encryption for Error Correction," SIAM AG, 2023.

6 **P. Yuan**, K. R. Duffy, M. Médard, "On the Role of Quantization of Soft Information in GRAND," SIAM-SEAS, 2023.

7 H. Sarieddeen, **P. Yuan**, K. R. Duffy, M. Médard, "Soft-input, soft-output joint data detection and GRAND: A performance and complexity analysis," ISIT, 2023.

8 J. Griffin, **P. Yuan**, P. Popovski, K. R. Duffy, M. Médard, "Code at the Receiver, Decode at the Sender: GRAND with Feedback," ITW, 2023.

9 **P. Yuan**, M. Coşkun, "Complexity-adaptive maximum-likelihood decoding of modified GN-coset codes," ITW, 2021.

10 **P. Yuan**, T. Prinz, G. Böcherer, O. İşcan, R. Böhnke, W. Xu, "Polar code construction for list decoding," SCC, 2019.

11 T. Wiegart, T. Prinz, F. Steiner, **P. Yuan**, "Design of polar codes for parallel channels with an average power constraint," ISIT, 2019.

12 **P. Yuan**, F. Steiner, "Construction and decoding algorithms for polar codes based on 2x2 non-binary kernels," ISTC, 2018.

13 T. Prinz, **P. Yuan**, "Successive cancellation list decoding of BMERA codes with application to higher-order modulation," ISTC, Hongkong, 2018.

14 **P. Yuan**, F. Steiner, T. Prinz, G. Böcherer, "Flexible IR-HARQ scheme for polar-coded modulation," WCNC, 2018.

15 T. Prinz, **P. Yuan**, G. Böcherer, O. İşcan, R. Böhnke, W. Xu, "Polar coded probabilistic amplitude shaping for short packets," SPAWC, 2017.

16 G. Böcherer, T. Prinz, **P. Yuan**, F. Steiner, "Efficient polar code construction for higher-order modulation," WCNC, 2017.

Patents

1 **P. Yuan**, M. Coşkun, Gerhard Kramer, "Decoding device and method for decoding," DE102020128918B3, Granted, Nov. 2021.

2 T. Prinz, T. Wiegart, **P. Yuan**, P. Schulte, Gerhard Kramer, O. İşcan, R. Böhnke, W. Xu, "Polar coding for parallel channels with different channel parameters," WO2021078389, Granted, Oct. 2019.

3 T. Prinz, **P. Yuan**, G. Böcherer, Gerhard Kramer, O. İşcan, R. Böhnke, W. Xu, "Polar coding with dynamic frozen bits," US20200052718, Granted, Apr. 2017.

4 **P. Yuan**, G. Böcherer, P. Schulte, Gerhard Kramer, R. Böhnke, W. Xu, "Error detection using symbol distribution in a system with distribution matching and probabilistic amplitude shaping," US10880037, Granted, Oct. 2016.

Computer Skills

C/C++, Julia, Python, Matlab

Languages

Mandarin	**Mothertongue**
English	**Fluent**
German	**Fluent**

References

Prof. Muriel Médard (medard@mit.edu)

Prof. Dr.sc.tech. Gerhard Kramer (gerhard.kramer@tum.de)

PD Dr.-Ing. habil. Georg Böcherer (georg.boecherer@tum.de)